

Estimating the dynamics of the groundwater in vegetated slopes based on the monitoring of streams



Kun Wang, Ze-Min Xu*, Lin Tian, Zhe Ren, Kui Yang, Yong-Jun Tang, Hai-Yan Gao, Jun-Yao Luo

Faculty of Civil Engineering and Mechanics, Kunming University of Science and Technology, Kunming 650500, China

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ABSTRACT

Rainfall is commonly considered to be responsible for landslides, but strictly, it is the groundwater recharged by rainfall that triggers slope failures. The dynamics of the groundwater in slopes is meaningful for assessing landslide hazard. Groundwater monitoring methods through boreholes or wells are effective but invasive, slow and expensive. In addition, these approaches are unsuitable for vast alpine environment prone to frequent landslides. To find an easy and inexpensive approach of estimating the dynamics of the groundwater in vegetated slopes, the response behavior of the stream flow to rainfall in the Touzhai valley (Southwestern China) is investigated based on in-situ monitoring. Combined with Complete Water Quality Analysis (CWQA) and a range of experiments (leaching test, in-situ infiltration and dye tracer infiltration), the source of the stream and its response mechanism to rainfall are determined. Some results are obtained: (1) the monitored stream flow fluctuates sensitively with the variation of rainfall without obvious hysteresis whether it is in the high- or low-water period; (2) the CWQA suggests that the stream water is derived from the slope groundwater rather than the direct rainfall or overland flow; (3) what leads to the stream flow rapid response is the macropore system (comprising of root-soil interstices, rotten root channels, inter-aggregate pores, worm pores and earth-gravel interstices) developed in the unsaturated zone of the slopes, which offers preferential conduits for rainfall infiltration. Therefore, monitoring streamflow rates can be an inexpensive and practical method to estimate the dynamics of groundwater in vegetated slopes, which, ulteriorly, provides the basis for diagnosing the stability of remote headwater slopes subject to failures. More than that, the alternative monitoring sites could be placed in inhabited valley bottoms or alluvial fans to avoid the absence of electric power and communication in uninhabited headwater catchments.

1. Introduction

In well-vegetated mountainous watersheds, intense and prolonged rainfall may not only induce flash floods in narrow valleys, but also trigger clustering slope failures in the vast valley walls, such as landslides and debris flows (e.g. Guzzetti et al., 1992; Godt, 1999; Cannon et al., 2001; Chigira, 2002; Guzzetti et al., 2004; Cardinali et al., 2006; Godt and Coe, 2007; Petley, 2012; Nikolopoulos et al., 2014; Vasu et al., 2016; Kumar et al., 2017; Piciullo et al., 2017; Saito et al., 2017). These catastrophic events have resulted in a large number of casualties and enormous economic damages worldwide (Nikolopoulos et al., 2015) and it is therefore important to predict their occurrences.

In general, empirical rainfall thresholds are used to judge whether a rainfall event would cause slope failures (e.g. Guzzetti et al., 2007 and Guzzetti et al., 2008; Brunetti et al., 2010; Rossi et al., 2012; Marra et al., 2014; Segoni et al., 2014; Wu et al., 2015; Ma et al., 2015;

Nikolopoulos et al., 2015; Marra et al., 2016; Saito et al., 2017). Owing to the presence of microclimates in the mountains (Church and Miles, 1987), rain gauges should be sufficiently close to potential landslide source areas (Wang et al., 2019). And this strategy often requires specialized power lines to operate the equipment and transmit data (Marchi et al., 2002), which are exactly lacking in uninhabited headwater catchments where slope failures occur mainly.

Furthermore, the trigger of vegetated slope failures is not rainfall itself in the strict sense, but the groundwater derived from rainfall. That is to say, landslides and debris flows following rainfall are in fact the stability response of slopes to groundwater (Dawes et al., 2004; Knight et al., 2005; Lana-Renault et al., 2007; Fiori et al., 2007; Mendicino et al., 2008). Therefore, monitoring the dynamic variation of the groundwater in slopes plays a direct and fundamental role in the studies of so-called “rainfall-induced landslides”. The main approach to monitor the groundwater in slopes is to rely on boreholes or wells with

* Corresponding author.

E-mail address: xzm768@kmust.edu.cn (Z.-M. Xu).

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or without piezometers. Banks et al. (2009) used a measurement well network with PVC piezometers to monitor the groundwater fluctuation processes in the soil, saprolite and fractured bedrock zones in the Mount Lofty Ranges, South Australia. Peng et al. (2010) investigated the source of the groundwater in a large slope and its influence on the creep of the slope in northern Taiwan based on two monitoring wells. Kosugi et al. (2011) conducted long-term hydrogeological observations using densely located bedrock wells in a headwater catchment with steep topography, located in the Rokko mountain range of southern Hyogo Prefecture, central Japan. Padilla et al. (2014) analyzed the groundwater response to rainfall of a hillslope affected by gravitational deformation based on the data obtained from two observation boreholes. Notwithstanding, monitoring the groundwater dynamics in slopes through boreholes is suitable for specific large landslides or slope deformations, and because of its being invasive, slow and expensive (Linde et al., 2007), application of such techniques becomes difficult in vast alpine areas prone to landslides without availability of electric power. More works are required to establish more practical methods in measuring the dynamics of the groundwater in slopes in steep and high mountains.

In well-vegetated alpine areas, rainfall intensity rarely exceeds the infiltration capacity of slopes (Reynolds and Thompson, 1988; Bonell, 1993; Sinha et al., 2016). Streams are recharged mainly by the groundwater in slopes which discharges in the forms of springs and drainages, and the contribution of overland flows to the streamflows is small (Sklash and Farvolden, 1979; Tanaka et al., 1988; Gabrielli et al., 2012). Mosley (1979) instrumented weirs to study the streamflow generation in a forested watershed in New Zealand and indicated that

subsurface flow in the soil was the predominant mechanism of channel stormflow generation. Gibson et al. (2000) found that shallow hillslope groundwater was a dominant source of the streamflow in sloping bog-forest uplands of the North Coast Forest District, British Columbia. Onda et al. (2006) studied the response of runoff to rainfall in high-relief and forest-covered mountainous watersheds and found that the subsurface stormflow was dominant even in these extremely steep mountains. Lana-Renault et al. (2007) demonstrated the subsurface flow was most likely to be the dominant runoff processes during wet conditions in a sub-Mediterranean catchment. Crespo et al. (2011) analyzed the rainfall-runoff relation of 13 intensively monitored micro-catchments, with an area ratio of vegetation cover > 74%, in the Andes of southern Ecuador and the results showed that streamflow primarily consisted of subsurface flow. Guzha et al. (2015) pointed out that shallow subsurface lateral pathways played a significant role in controlling runoff generation processes in a forest catchment in the Amazonia region.

The published work suggests that the information on the dynamics of the groundwater in vegetated slopes, especially on the groundwater level fluctuation may be indirectly estimated by means of monitoring stream flows. What's more, the monitoring of stream flows may be performed in inhabited valley bottoms or depositional fans downstream of unstable slopes, where the infrastructures including power and communication are generally perfect. In the meantime these lower channels and fans themselves are potentially affected areas by upstream slope failures. Therefore, from a practical point of view, estimating the dynamics of the groundwater in vegetated slopes based on the monitoring of stream flow merits in-depth investigations.

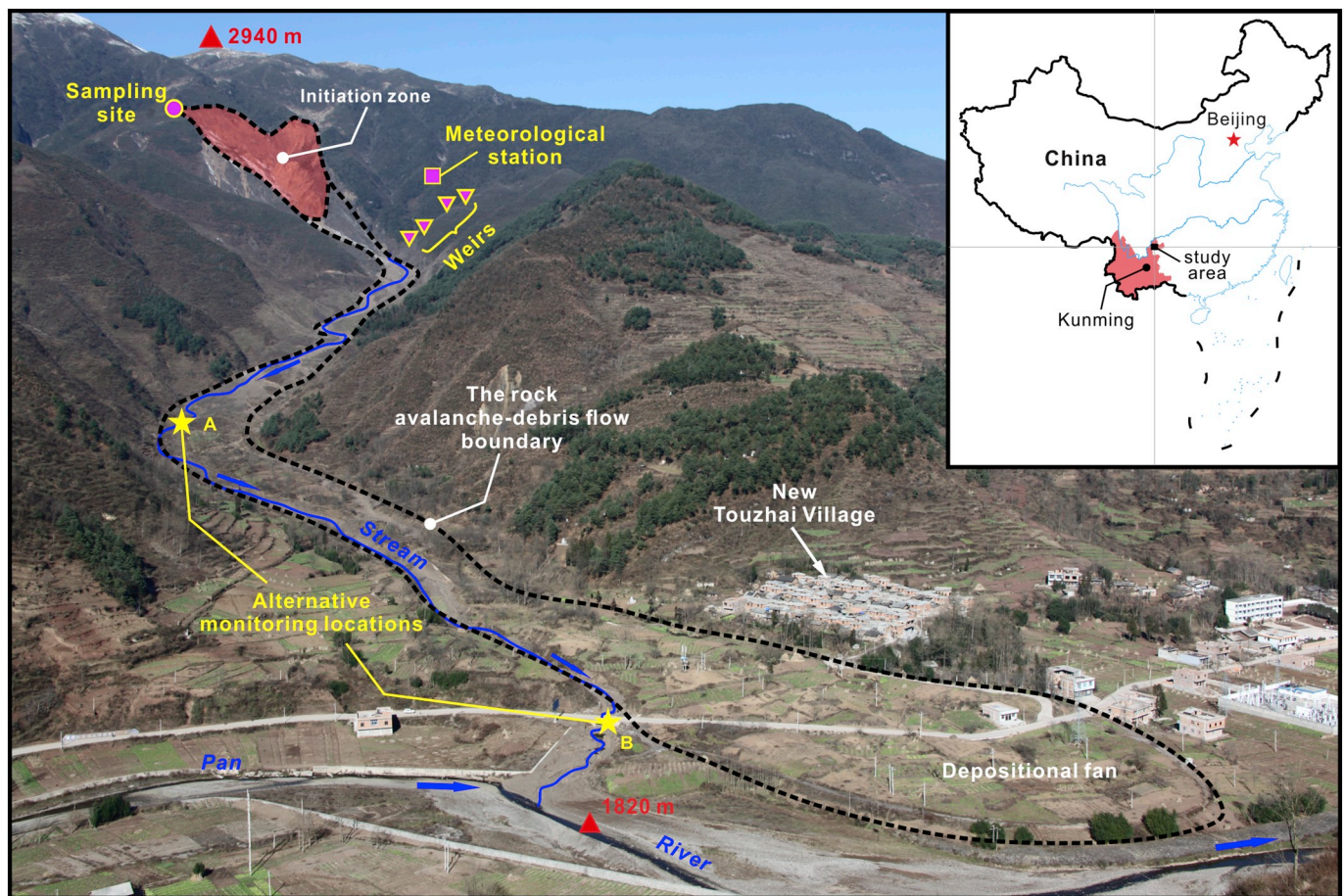


Fig. 1. The panorama of the study area and its geographical location, where a massive landslide with a volume of about 9 million m^3 (the Touzhai rock avalanche) occurred on September 23, 1991. The rock avalanche-debris flow buried the Touzhai village distributed along the former valley bottom and depositional fan, killing 216 people (Ren et al., 2018). View is to the northwest.

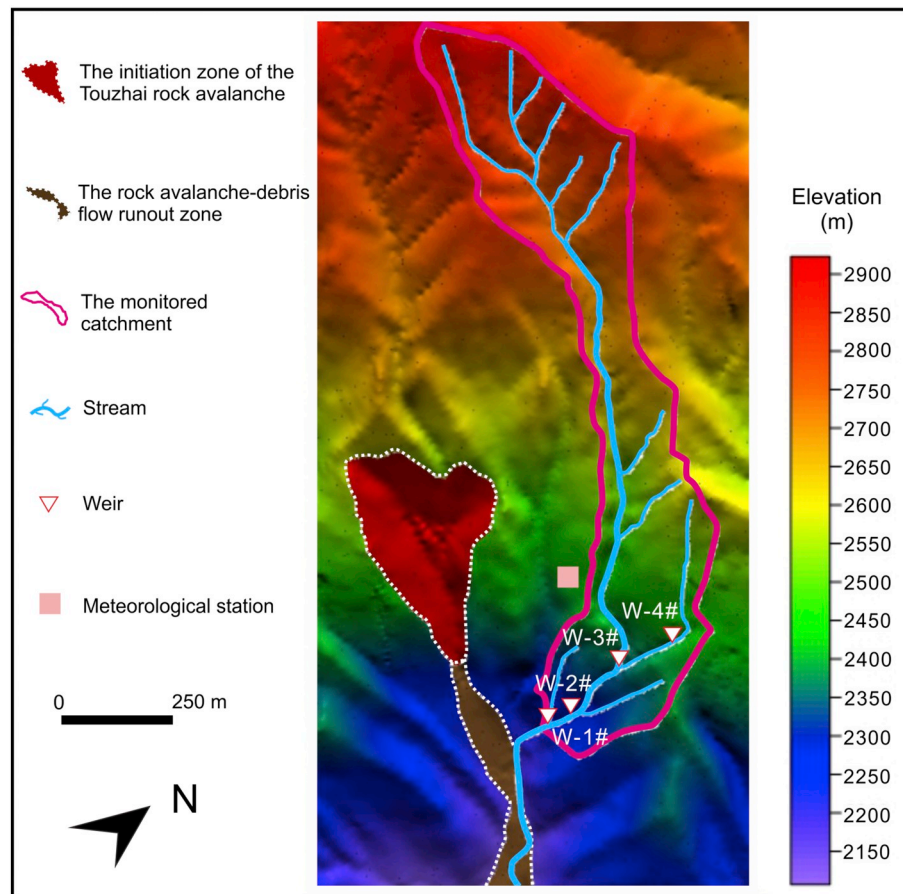


Fig. 2. Map showing the monitored catchment on the left side (facing downslope) of the Touzhai rock avalanche initiation zone.

Here we present a case example linking streams with the groundwater in vegetated slopes. Using 1 year of stream discharge and rainfall data measured in a forested slope area at the head of the Touzhai Valley, Yunnan, China (Fig. 1), we show how the stream flow responds rapidly to rainfall processes. Our further investigation and explanation of the source of the stream and its rapid response mechanism to rainfall uniquely incorporate a broad range of analyses and tests, including the Complete Water Quality Analysis (CWQA), leaching experiments, in-situ infiltration experiments and dye tracer infiltration experiments. The results that the dynamics of the groundwater in vegetated slopes can be qualitatively estimated through the streams downstream of the unstable slopes and the estimation may act as a diagnostic feature of vegetated slope stability are applicable to other forested catchments worldwide.

2. Site description

The Touzhai valley of about 2.5 km long lies ~300 km northeast of Kunming and is heavily vegetated. The drainage area is approximately 3.2 km² and the elevations in the area range from 1820 (at the Pan River) to 2940 m a.s.l. (at the main divide) (Fig. 1).

Geologically, the Touzhai valley is composed by the rocks belonging to the west limb of the Pan River (Fig. 1) syncline striking north-northeast as lower Triassic sandstones and siltites of the Feixianguan formation (T_{1f}), upper Permian sandstones, conglomerates and shales of the Xuanwei formation (P_{2x}) and upper Permian basalt of the Emeishan formation (P_{2β}). In more detail, the valley is located in a monocline striking NE–SW and dipping 25°–40° to the SE. The Triassic–Permian terrigenous clastic sedimentary rocks are widely exposed on the valley walls of the most reaches, and P_{2β} is mainly distributed at the head of the main valley.

As a tributary of the main stem of the Touzhai Creek, the monitored small catchment in this study (Figs. 1 and 2 and the monitoring method in Section 3.1) is in the eastern part of the valley head, and covered by eluvial organic-rich deposit on basalt bedrock to a depth of up to 1.5 m. The distinguished basalt types include aphyric basalt, amygdaloidal basalt and porphyric basalt with plagioclase phenocrysts.

According to our monitoring, in the upstream valley from October 2013 to September 2014, the maximum, minimum and average temperatures were 27.1 °C, −19.3 °C and 8.5 °C, respectively. The total rainfall was 1726.8 mm with an average of 143.9 mm/month, maximum of 192.4 mm (August) and minimum of 31.6 mm (December). The rainfall in wet season (from May to September) was 1252.4 mm, accounting for 73.5% of the whole year's rainfall (Wang et al., 2019). A perennial stream develops along the Touzhai Valley (Fig. 1).

The Touzhai valley is covered by a natural forest with a coverage proportion of > 95%. The field investigations identify 31 species of plants in the forest, which mainly belong to the families of *Rosaceae*, *Ericaceae*, *Asteraceae* and *Salicaceae*. Furthermore, the main soil animal is earthworm, which belongs to , *Megascolecidae Pheretima*.

3. The dynamic response of the stream flow to rainfall

3.1. The monitoring method

The upstream slope area on the left side (facing downslope) of the Touzhai valley, about 500 m to the initiation zone of the rock avalanche, is selected as the stream monitoring area. Four weirs, numbered W-1#, W-2#, W-3# and W-4# according to their altitudes from low to high, are set to quantify the volume flow-rates in the creek channels (Figs. 1–3). Their controlled drainage areas are 0.40 km² (W-1#), 0.37 km² (W-2#), 0.30 km² (W-3#) and 0.03 km² (W-4#), respectively.

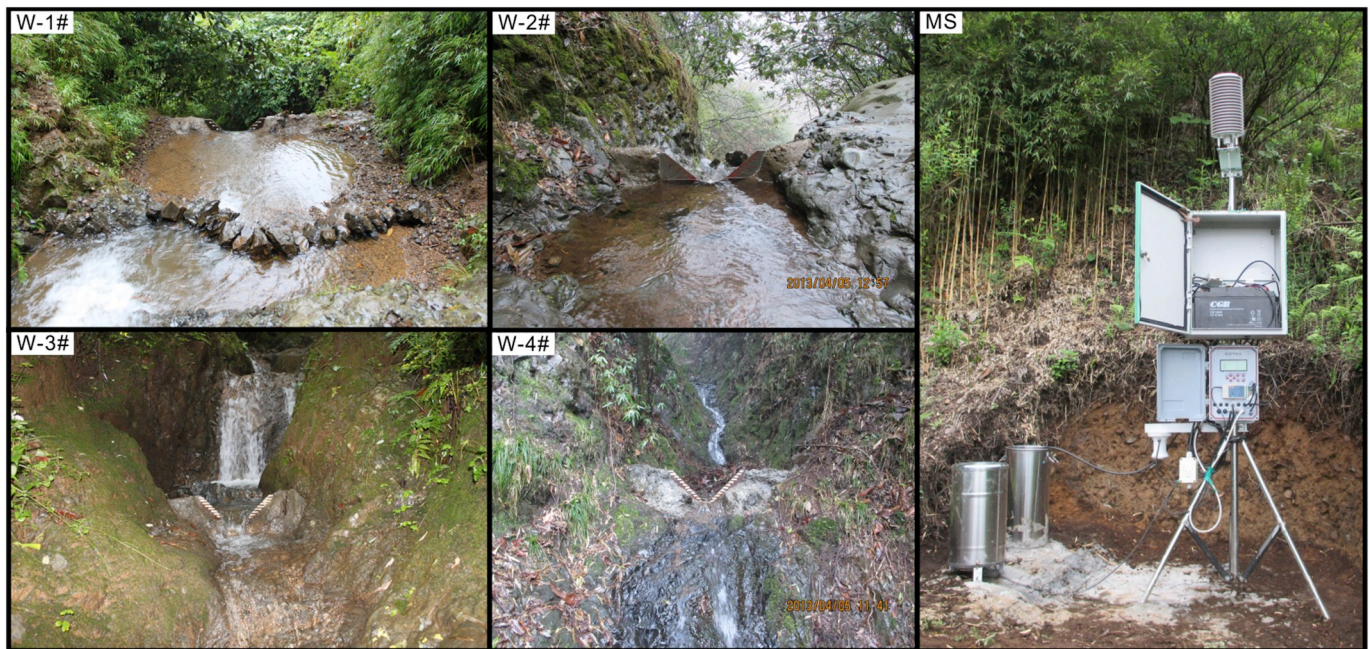


Fig. 3. The four weirs and the automatic Meteorological Station (the original photo of Meteorological Station can be found in Wang et al. (2019)). Their locations are marked in Figs. 1 and 2.

The water levels behind the weirs are recorded weekly and the obtained and corrected data are converted to Flow Rate (FR) according to the equation below (BS ISO 1438, 2008).

$$Q = Ah^{\frac{5}{2}}$$

where Q is the water FR (m^3/s), h is the water level (m), and coefficient A is determined according to BS ISO 1438 (2008).

An automatic Meteorological Station (MS) with a monitoring resolution of 0.2 mm is installed beside the monitored small catchment to evaluate the response behaviors of the stream FR to rainfall (Figs. 1–3). The MS is located at the altitude of 2400 m a.s.l and driven by two 64 V storage battery, which are rotated every week. The recorded rainfall data are transmitted in real time to the specified receiver in Kunming (Fig. 1) using GPRS.

The monitoring period, both of the stream FR and rainfall amount, is from September 2013 through September 2014.

3.2. Results and analyses

3.2.1. The average stream FR and the rainfall amount at the monthly scale

As stated in Section 3.1, stream FR values of the four weirs are recorded once a week. Thus, there are 4 or 5 FR data points for each weir in a month, and their averages, which are used to represent FR in the current month, are depicted in Fig. 4, along with the corresponding monthly rainfall amount.

The monthly mean FRs measured by the four weirs followed the relationship of $\text{FR}_{\text{W-1\#}} > \text{FR}_{\text{W-2\#}} > \text{FR}_{\text{W-3\#}} > \text{FR}_{\text{W-4\#}}$, which was in accordance with their controlled drainage areas.

At the monthly scale, the measured FR values of the four weirs varied considerably but showed a good agreement with one another (Fig. 4). Based on the monthly FR values, the 12 months can be divided into three water periods, i.e. the high-water period (from June to September), the low-water period (from December to the next March) and the normal-water period (between the former two). The separate maximum FR values of the four weirs in the high-water period were 17.7, 27.7, 80.5 and 16.25 times more than their minimum FR values in the low-water period.

In the monitoring period, a total of 1726.8 mm rainfall amount was

detected by the MS, with the minimum of 31.6 mm in November, maximum of 392.4 mm in August and average of 143.9 mm/month. The four stream FRs were highly consistent with the fluctuation of monthly rainfall, with the Pearson Correlation Coefficients (PCC henceforth, Pearson, 1895) of 0.953 (W-1#), 0.965 (W-2#), 0.929 (W-3#) and 0.949 (W-4#), which indicated that stream FR could reflect the variation of monthly rainfall amount.

3.2.2. The dynamic variation of stream FR and the rainfall amount at the weekly scale

Compared to the situation at the monthly scale, the fluctuation of the four stream FR values at the weekly scale also synchronized with one another, but more violent. The 12 months may be simply divided into two periods, i.e. the high-water period, from May to September, and the low-water period, from October to the next April (Fig. 5). The respective average FR values of the four weirs in the high-water period were 5.8, 6.7, 8.6 and 4.3 times as large as their average FR values in the low-water period.

In the high-water period, all of the FR values in the four weirs varied sharply and the individual maximum FR values of W-1#, W-2#, W-3# and W-4# were 19.5, 23.4, 26.5 and 8.9 times over their minimum FR values. More importantly, the FR of each weir changed rapidly, and all of the increase-peak-decrease processes occurred within 1 week.

Although the peak FR values measured in the low-water period were much less than those measured in the high-water period, the FR fluctuation of each weir was also drastic. The ratios of the respective maximum FR values to their minimum FR values in W-1#, W-2#, W-3# and W-4# were 18.7, 30.5, 286.4 and 130.6. Especially, the FR values of W-3# and W-4# in January and February 2014 almost descended to zero, with a minimum value of 0.027 L/s for W-3# and 0.014 L/s for W-4# on February 2nd, 2014.

The rapid response of the stream FR to rainfall is more pronounced over the weekly scale. All of the six FR peak values in the high-water period appeared during the rainfall processes with intense storms, and the FRs decreased quickly with the end of rainfall processes. The FR was also sensitive to rainfall in the low-water period, when the FR values were relatively small. For instance (the dotted rectangle in Fig. 5), as the prior weekly cumulative rainfall amount before 2014/03/22 was only 11.6 mm, the FRs of the four weirs on 2014/03/22 were all at a

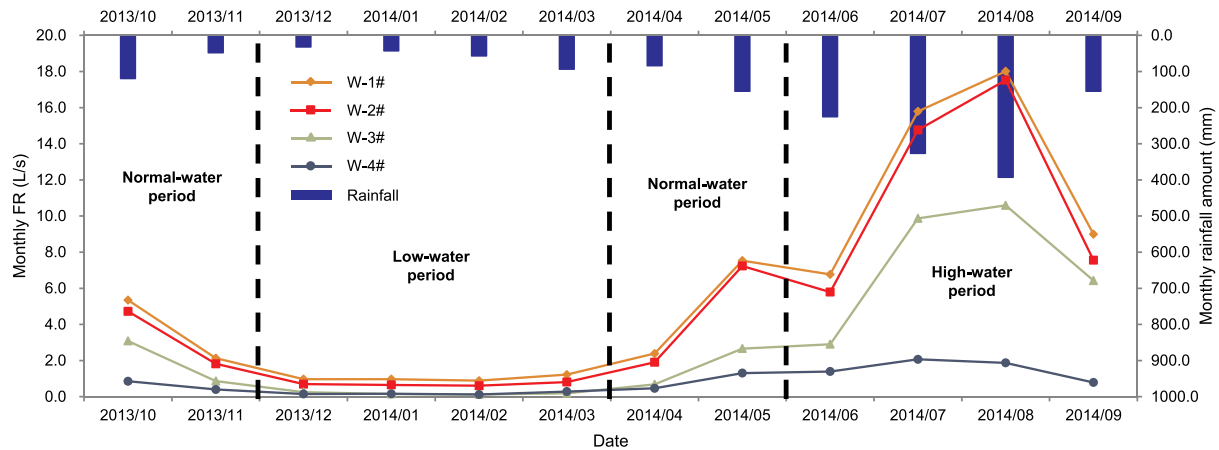


Fig. 4. Average monthly stream FRs of the four weirs (W-1# to W-4#) and the corresponding monthly rainfall amount (the rainfall data after Wang et al. (2019)).

very low level, with the maximum of only 0.794 L/s for W-1#. Accompanying the continuous rainfall of 40, 39.6 and 36.6 mm for the next 3 weeks (2014/03/23–2014/04/12), all of the FRs in the four weirs began to increase and peaked during this period. In the following weeks, rainfall disappeared (from 2014/04/13 to 2014/04/19), and the FRs decreased rapidly. The stream was close to dry up.

In summary, at the weekly scale, the stream FR values vary almost synchronously with the rainfall and exhibit a high indication accuracy for rainfall.

3.2.3. The dynamic variation of stream FR and the rainfall amount at the daily scale

To further investigate the indication of streams on rainfall, the stream FR measurement once a day was performed from September 4th, 2013 to September 16th, 2013 and the results (Fig. 6) showed that FRs of the four weirs responded sensitively to the rainfall, even to the daily rainfall amount of only a few millimeters.

Corresponding to the rainfall of 3.8 and 3.6 mm on September 4th and 5th, 2013, the stream FRs of the four weirs gradually increased and separately reached to their peaks, followed by a gradual decrement until September 9th. After the rainfall on September 10th, the stream FRs were elevated by the intense rainfall of 28.6 mm on September 11th to the highest peak in this monitoring period with high resolution. Because the precipitations were relatively low on September 12th and

13th, they could not prevent the stream FRs to decline.

The results presented above show that the stream FR fluctuates sensitively with the variation of rainfall without obvious lag time whether it is in the high- or low-water period. This conveys that if the stream water is derived from the groundwater, the information on the dynamics of the groundwater following rainfall in vegetated slopes can be delivered by stream water tables. Postulating this is tenable, the monitoring locations of stream flows could be placed in the inhabited areas, such as the alternative locations marked in Fig. 1, to avoid the problems of lacking electric power and communication in headwater catchments.

4. The source of the stream water

To determine the source of the stream water, the chemical compositions of the stream water, rainwater and laboratory leachates were carefully examined.

Two bottles of stream water were taken once per month, giving a total of 14 sampling days (28 bottles) from May 2013 to June 2014. Two buckets of rainwater samples were collected from late May to the early June 2013. The soil samples used for the leaching experiment were sampled from the right flank (facing downslope) of the rock avalanche initiation zone (Fig. 1). Three layers from up to down were taken in the vertical direction: humus (S₁), clay (S₂) and heavily weathered basalt (S₃).

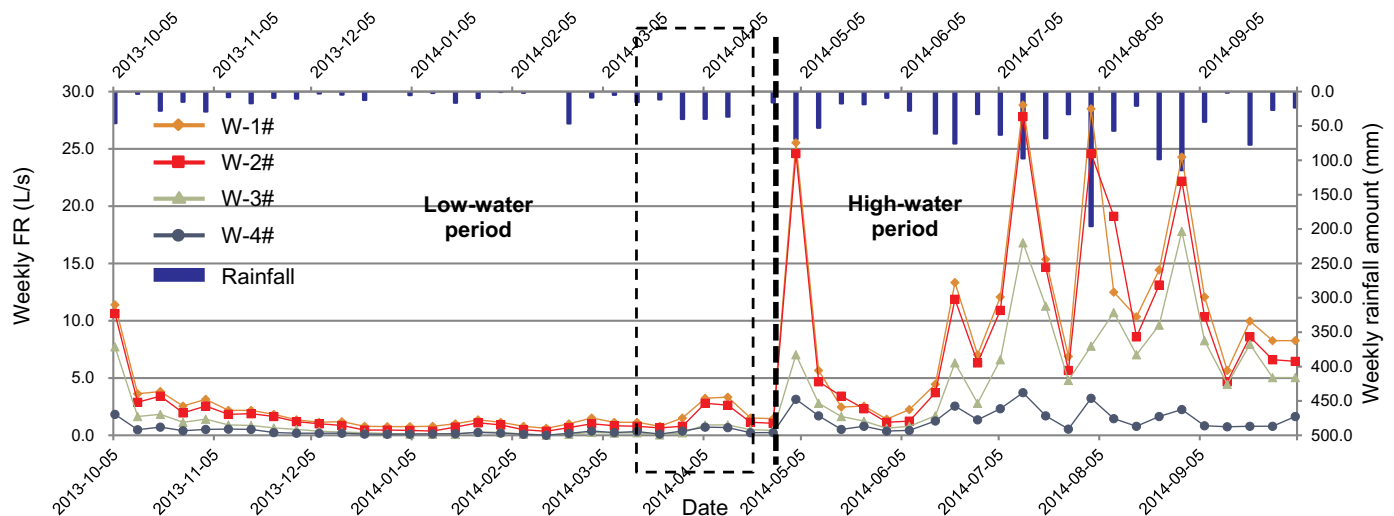


Fig. 5. Time-trends of the weekly stream discharge of the four weirs (W-1# to W-4#) and the corresponding weekly rainfall amount (the rainfall data after Wang et al. (2019)). The zone constrained by the dotted rectangle highlights the rapid response of stream FR to rainfall in the low-water period.

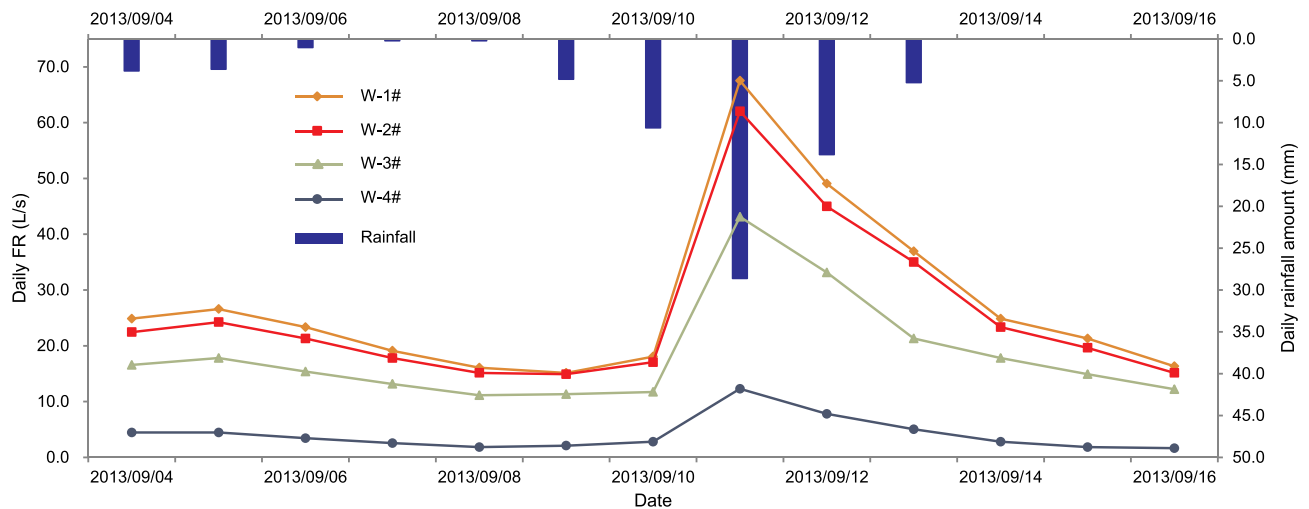


Fig. 6. Volume flow-rate of the four weirs and corresponding daily rainfall amount (the rainfall data after Wang et al. (2019)) at the daily scale as functions of time. The measurement time of the flows was 9: 00–10:00 every morning.

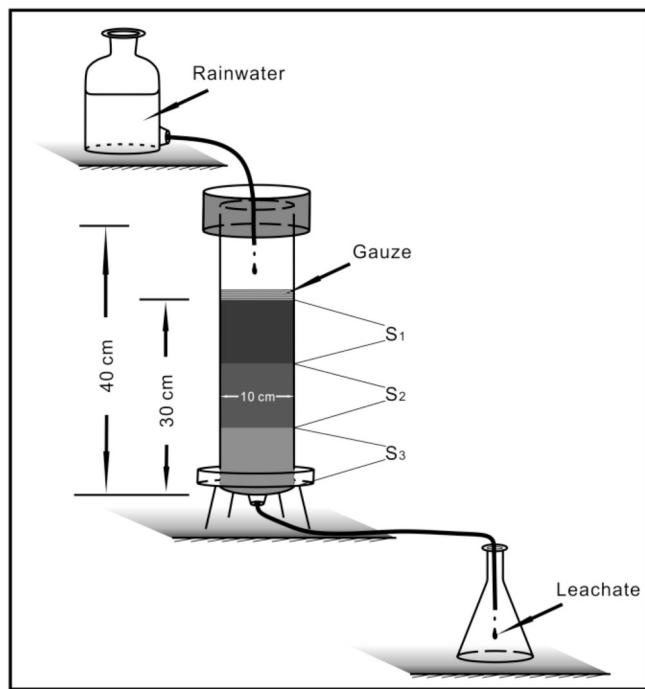


Fig. 7. The leaching experimental apparatus. S₁, S₂ and S₃ refer to humus, clay and heavily weathered basalt.

The leaching experiment apparatus is shown in Fig. 7. The three soil samples (S₁, S₂ and S₃) were compacted into a tailored bucket with an inner diameter of 10 cm and height of 40 cm. From bottom to top, the thicknesses of S₃, S₂ and S₁ were all 10 cm, which formed a 30 cm high column of soil. The rainwater was input to the surface of the column, which was covered with gauze. A conical flask was placed below the bucket to collect the leachate. The leaching experiment was repeated five times, and the gained leachates were numbered L₁, L₂, L₃, L₄ and L₅.

The CWQA for the stream water, rainwater and leachates were performed to investigate their chemical compositions.

4.1. Chemical compositions of the stream water, rainwater and leachate

The relative concentrations of major ions of the stream water,

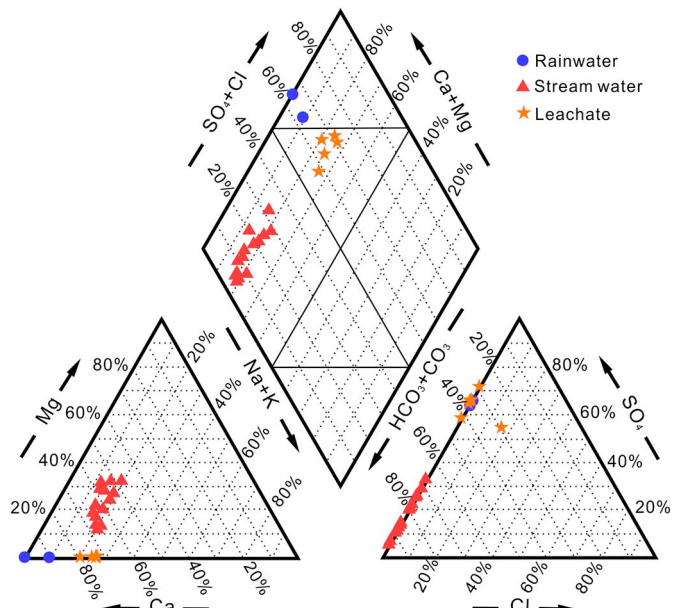


Fig. 8. Piper trilinear diagram for the major ion relative concentrations of the stream water, rainwater and leachate.

rainwater and leachate are plotted in the Piper trilinear diagram (Piper, 1944) (Fig. 8). The plot of chemical data in the central diamond field indicates that these water samples can be classified into different hydrochemical types, i.e., HCO₃—Ca—Mg type for the stream water samples, SO₄—Ca—Mg type for the two rainwater samples and SO₄—Cl—Ca—Mg type for the five leachate samples. In the case of cations, all of the water samples are dominated by the alkaline earths (Ca²⁺ and Mg²⁺) over alkalis (Na⁺ and K⁺). In the case of anions, the dominance of strong acids (SO₄²⁻ and Cl⁻) exceeds weak acids (HCO₃⁻ and CO₃²⁻) for the rainwater and leachate samples, and the inverse occurs in the stream water samples.

From the above results, the difference in chemical composition among the three types of water samples is mainly concentrated on anions, especially the SO₄²⁻ and HCO₃⁻ (the anion triangle field in Fig. 8). The average concentrations of anions (SO₄²⁻ and HCO₃⁻), as well as other composition parameters (pH, silicon dioxide (SiO₂) and total dissolved solids (TDS)), are illustrated in Fig. 9.

The average pH of the stream water samples is 7.77. The average

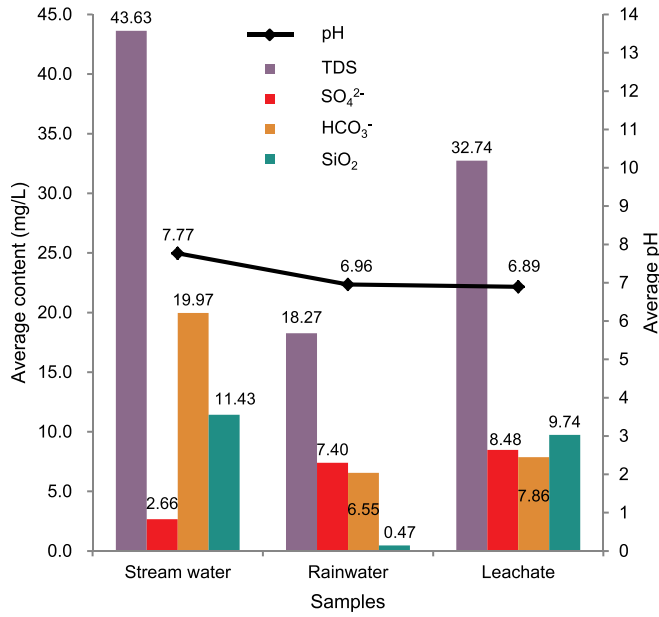


Fig. 9. Comparison of the pH, total dissolved solid (TDS) and concentrations of the major constituents between the stream water, rainwater and leachate.

contents of SiO₂ and HCO₃⁻ are 11.43 and 19.97 mg/L, respectively. The SO₄²⁻ content is low with a mean of 2.66 mg/L. The average TDS is 43.63 mg/L, which is the highest in the three types of the water samples.

Compared with the stream water, the two rainwater samples are neutral with an average pH of 6.96. The anions are mainly SO₄²⁻ and HCO₃⁻ with an average contents of 7.40 and 6.55 mg/L, respectively. The SiO₂ content is very low with an average of only 0.47 mg/L. The TDS averages 18.27 mg/L. The chemical composition of rainwater indicates that it is characterized by a low degree of mineralization in an oxidizing environment.

The pH values of the five leachate samples range from 6.70 to 7.20, with an average of 6.89. The SiO₂ content in the leachate was slightly lower than that in the stream water, but much higher than that in the rainwater. Similarly, the mean content of HCO₃⁻ was lower than that in the stream water, but higher than that in the rainwater. The SO₄²⁻ content was the highest of the three types of water samples. TDS displayed the same trend as HCO₃⁻.

4.2. Discussions of the stream generation

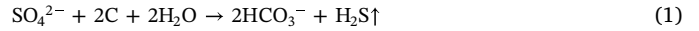
The detailed geomorphologic reconnaissance of the monitored catchment (Fig. 2) with a precision of 3 m × 3 m could not find rills and other signs of overland flow activity. Therefore, the stream water is not the rainfall concentrated at the surface.

There should be hydrochemical signatures about the stream generation in the above analysis results. Fig. 9 shows that the TDS of the stream water is higher than that of the leachate, and significantly elevated above that of rainwater, indicating that the stream flow should be derived from the groundwater.

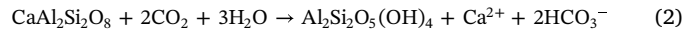
From the PCC matrix showing the relationship between chemical constituents in the stream water (Table 1), the following correlations can be found among the ions: SO₄²⁻ shows a strong negative correlation with SiO₂ (-0.80), Na⁺ (-0.68), Mg²⁺ (-0.60) and HCO₃⁻ (-0.68); HCO₃⁻ is highly positively correlated with SiO₂ (0.76), Na⁺ (0.81), Ca²⁺ (0.70) and Mg²⁺ (0.72); and SiO₂ has a good correlation with Na⁺ (0.80) and Mg²⁺ (0.63). These correlations, combined with the environment of the slope soil, allow us to infer the hydrogeochemical processes with respect to SO₄²⁻, HCO₃⁻ and SiO₂.

4.2.1. SO₄²⁻ and HCO₃⁻

The SO₄²⁻ in water can be reduced to H₂S by desulfurizing bacteria inside the slope soil mass, where it is the anoxic reducing environment with numerous organic matters. The reaction is as follows.



Besides, the HCO₃⁻ can be also generated in other ways. In magmatic landscapes, the increment of HCO₃⁻ concentration can be caused by the kaolinization of feldspar (Tay et al., 2014; Ma, 2016; Yang et al., 2017).



The feldspar in the basalt can be dissolved in the CO₂ and H₂O environment, generating abundant HCO₃⁻. This reaction is corroborated by the micro structure of the basalt-saprolite blocks wrapped in the unsaturated soil mass in the Touzhai valley. As shown in Fig. 10a, long strip-shaped microcrystals of plagioclase in the basalt saprolite are intensely dissolved universally. This makes most fields of microscope view under single polarized light to appear pink (the color of the resin as filling material) with a surface porosity of up to 8%. Irregular solution pores filled with pink resin can be identified in some relatively large plagioclase crystals (Fig. 10b and c).

The preceding chemical processes will lead to the decrement of the SO₄²⁻ content and the increment of the HCO₃⁻ content and pH, which coincides with the variations of the corresponding constituents from the rainwater to stream water (Fig. 9). However, it is recognized that the concentrations of SO₄²⁻ and HCO₃⁻ in the leachate do not follow this rule. This is due to the fact that no enough time is provided for the performance of reaction (1) and (2) in the leaching experiment.

The analysis of the concentration differences of SO₄²⁻, together with the HCO₃⁻ in the three types of water, indicates to some extent that the stream water can be regarded as the leachate formed by rainfall leaching the slopes.

4.2.2. SiO₂

The generation of SiO₂ may be due to the dissolution and chloritization of pyroxene (Yang et al., 2017).

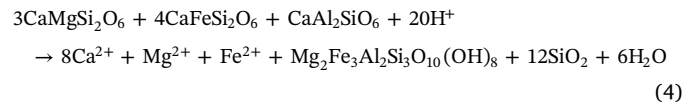
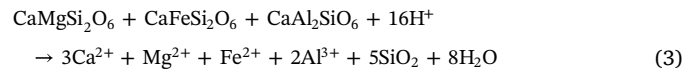


Fig. 9 shows that the average concentrations of SiO₂ in stream water and leachate are 11.43 and 9.74 mg/L, which are 24.3 and 20.7 times as high as that in rainwater, respectively. The sharp rise of SiO₂ from rainwater to stream water and leachate is in accordance with that of TDS mentioned above. As a summary, the high TDS and the similar magnitude of SiO₂ concentrations in the stream water and leachate suggest that the stream water beside the initiation zone of the Touzhai rock avalanche is derived from the groundwater rather than the direct rainfall or overland flow (similar conclusion can be found in Walsh and Howells, 1988). Thus, the rapid variation of the stream FR with rainfall stated in Section 3, in fact, reflects the real-time fluctuation of the groundwater table in the slopes following rainfall.

5. The mechanism of the rapid response of the groundwater to rainfall

Based on the works stated in Sections 3 and 4, the groundwater is identified as the source of the stream water. Transformation mechanism from rainfall to groundwater to stream water is investigated in this section. First, the infiltration rate of the slope soil at different depths (0, 10 and 60 cm) is measured by in-situ double-ring infiltration

Table 1

Pearson Correlation Coefficients (PCC) of chemical parameters in the stream water. The bold values highlight the strong correlations between different chemical constituents.

Ions	SiO ₂	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	SO ₄ ²⁻	HCO ₃ ⁻	CO ₂	TDS
SiO ₂	1.00								
Na ⁺	0.80	1.00							
K ⁺	0.34	0.35	1.00						
Ca ²⁺	0.40	0.76	0.16	1.00					
Mg ²⁺	0.63	0.59	-0.01	0.42	1.00				
SO ₄ ²⁻	-0.80	-0.68	-0.22	-0.31	-0.60	1.00			
HCO ₃ ⁻	0.76	0.81	0.14	0.70	0.72	-0.68	1.00		
CO ₂	-0.18	-0.18	0.08	-0.00	-0.02	-0.14	0.19	1.00	
TDS	0.80	0.84	0.14	0.71	0.65	-0.60	0.90	-0.07	1.00

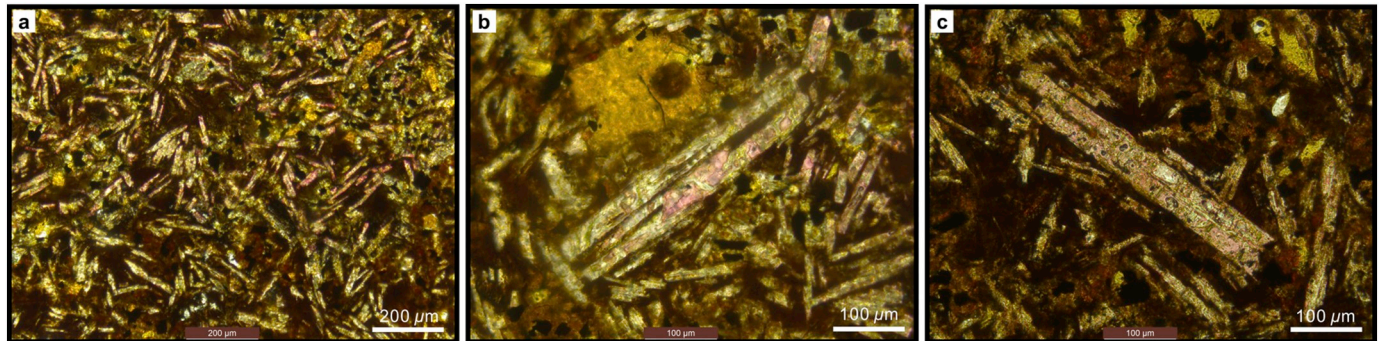


Fig. 10. The micro structure of basalt saprolite under a polarizing microscope. The pores left behind by the dissolution of plagioclase are occupied by pink resin. (a) the dense plagioclase microcrystals dissolved partly; (b) and (c) long strip-shaped microcrystals of plagioclases with intragranular micropores. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

experiment (Section 5.1). Then, the field dye tracer infiltration experiment is performed to identify the factors operating in the rapid response of the groundwater, and the subsequent stream water to rainfall (Section 5.2).

5.1. The hydraulic conductivity of the slope soil mass

5.1.1. The experimental approach

Single or double ring infiltrometers are commonly used for in-situ measurements of infiltration rate (e.g. Touma and Alberge, 1992; Gregory et al., 2005; Fatehnia et al., 2016; Morbidelli et al., 2017), but the double ring infiltrometer is the preferred choice, in which the use of the outer ring helps to reduce the lateral flow in soils (Bouwer, 1986). The experiment was conducted in two different sites, numbering as P-1# and P-2#, and subsequent calculation of soil infiltration rate was carried out following ASTM (2009).

5.1.2. The experimental result

Fig. 11 shows the variations of soil infiltration rate with time elapsing. In P-1#, the infiltration process lasted for 120 min, which could be divided into three stages: fluctuant declining stage from 0 to 55 min, slow declining stage from 55 to 90 min and stable stage from 90 to 120 min. In contrast, the water infiltration curve of P-2# displayed a different pattern compared to that of P-1# and could be separated into four stages. It started with a rapid declining stage, from 0 to 10 min. The fluctuant declining stage only lasted for 10 min. The slow declining stage, lasting for 70 min, was much longer than that of P-1#. What was consistent was that both of the two curves stepped into the stable stage after about 90 min.

The average values of infiltration rate in the stable stage can be regarded as saturated hydraulic conductivity (K_s) (ASTM, 2009). The K_s of the two sites are shown in Fig. 12. For P-1#, the K_s at the depths of 0, 10 and 60 cm are respectively 3.11×10^{-2} , 1.15×10^{-2} and 1.77×10^{-2} cm/s, of which the variation with depth is not significant.

The according K_s of P-2# are 7.74×10^{-3} , 3.58×10^{-3} and 5.02×10^{-3} cm/s, which show a similar pattern with P-1#. The K_s values measured at three depths in each site are within the same order of magnitude, and the reproducibility of the permeability experiment is demonstrated to be good by this result. The difference of 1 order of magnitude in K_s between the two locations should be understandable, especially in terms of the inherent heterogeneity of slope soils.

The average K_s of the two sites at the three depths are equal to 1.94×10^{-2} , 7.54×10^{-3} and 1.13×10^{-2} cm/s, respectively, which are in agreement with the hydraulic conductivities of forested soils reported in other studies. For example, Dunne (1978) indicated that the infiltration rate of the well-forested soil is generally above 2.22×10^{-3} cm/s. Likewise, Wang et al. (2000) found that the saturated infiltration rate of the forest soil in Hokkaido, Japan was 1.15×10^{-2} cm/s, which was relative higher than that of any other soil types, such as grass soil, tillage soil and bare land soil. These results show the tested soils in this case are very pervious (Bear, 1972) and it is the higher permeability of the slope soil that allows the groundwater and resultant streamflow to response rapidly to rainfall. Next, the water infiltration details in the slope soils will be revealed.

5.2. The pores in the slope soil mass

5.2.1. The design of the in-situ dye experiments

The field dye tracer infiltration experiments were conducted in April 2012 (low-water period) and September 2013 (high-water period). A square wooden frame with a side length of 1.0 m was pounded into the soils. 200 L solution of methylene blue or rhodamine B, either of which is visible in soils and has a low toxicity, a high stability over a large pH and ionic strength range (Zhang and Xu, 2016), was sprinkled to the slope surface of each test site in 4 h to imitate rainfall intensity of 50 mm/h, which is equivalent to extremely torrential rains (> 10.42 mm/h) according to rainfall grade classification (Wang et al., 2019). However, no overland flow was observed during the period of

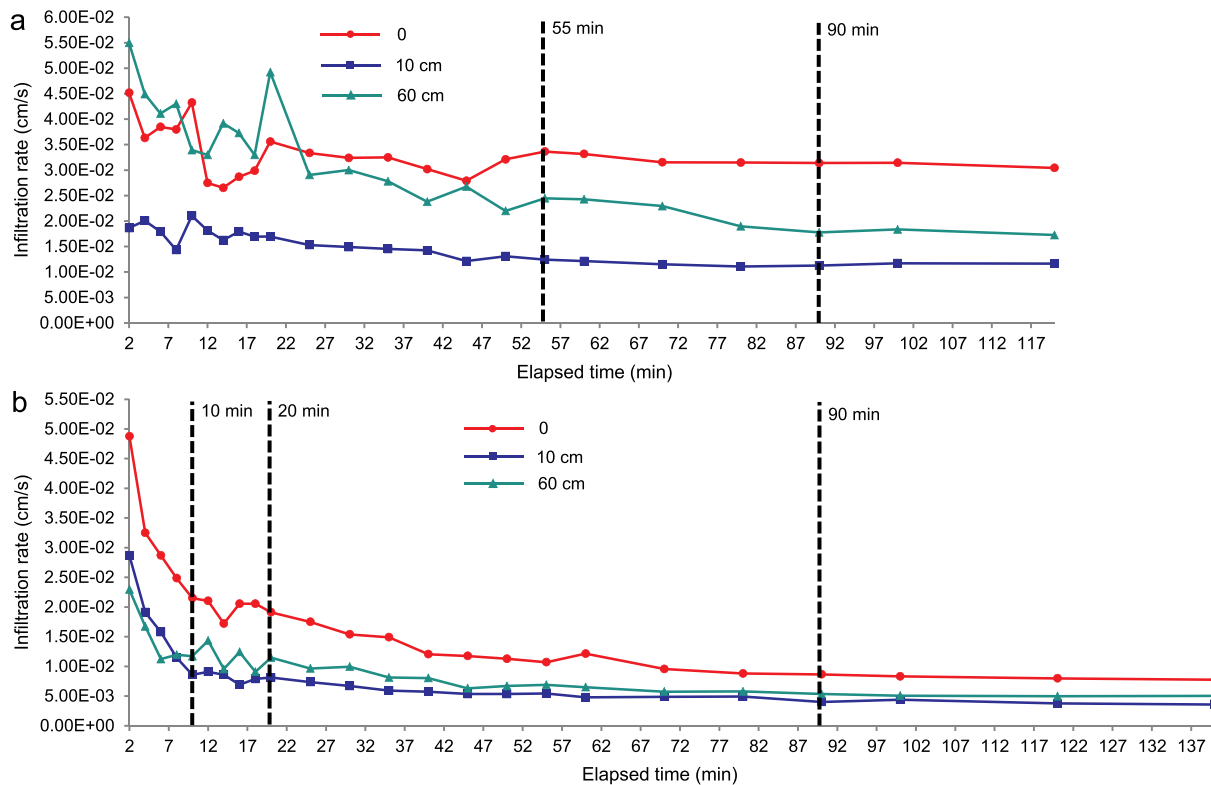


Fig. 11. Water infiltration curves of the two experimental pits at different depths. (a) refers to P-1# and (b) refers to P-2#.

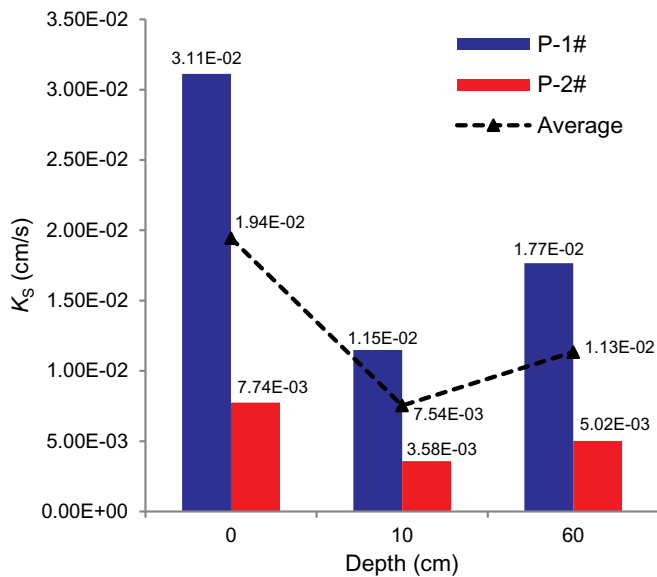


Fig. 12. Saturated hydraulic conductivity (K_s) versus depth for P-1# and P-1# test pits.

sprinkling, implying that the solution was totally absorbed by the soil masses. Then the sprinkled areas were sealed with cling films to prevent evaporation and left for 24 h to promote infiltration. Finally, a sequence of observation profiles were excavated and trimmed with a scissor and a brush. The detailed schematic diagram for this experiment can be found in Zhang and Xu (2016).

5.2.2. Results and discussions

The excavations (Fig. 13a) revealed that the dyed regions were mainly concentrated at 0–30 cm depth. In the depth range of

30–100 cm, the dyed area decreased dramatically and appeared only in macropores (Fig. 13a-iii and iv). Some dyed macropores had gone deep into the depth of 1 m or more, where the basalt bedrock was found. Carefully observations suggested that the infiltration of the solution was regulated overwhelming by macropores, which included root–soil interstices, rotten root channels, inter-aggregate pores, worm pores and earth–gravel interstices.

5.2.2.1. Root–soil interstices. As mentioned above, the exposed dyed regions were mainly in the depth range from 0 to 30 cm. This is due to the large numbers of roots derived from short shrubs or herbaceous plants (Fig. 13b). Two measurement sites of the root volume density (A-1# and A-2#) near the dyeing pits showed that it sharply decreased at the depth of 30 cm (Fig. 14). The density at the depth of 0–20 cm was 16,961.25 g/m³ for A-1# and 14,082.5 g/m³ for A-2#, with an average value of 15,521.88 g/m³, which was much higher than that at any other depth range. The density of root system is so high that the soil mass hosting roots becomes looser and more porous. The pervasively dyed root surfaces and adjacent soil walls (Fig. 13b-ii and iii) suggest that the root–soil interstices act as important pathways in rainfall infiltration.

5.2.2.2. Rotten root channels. Some dead roots will decay with time elapsing and leave a thick, tough, resistant bark, in which channels will form (Aubertin, 1971). On the experimental profiles, consecutive thready dyed traces were scattered, some of which had already extended to the depth of 80–100 cm (Fig. 13a-iii and iv, Fig. 13c-i). They were mainly originated from the decomposition of roots, displaying the tubular and dendritic shapes. Most of them were empty, with the residual bark surrounded (Fig. 13c-ii), and some were partly filled with the subsequent new fibrous roots (Fig. 13c-i). Their diameters were various, ranging from 1 to 5 mm, which was determined by the diameters of original roots (Fig. 13c).

In addition, Fig. 13c also conveys that the soil matrix near the rotten root channels is not dyed, indicating that these channels play a crucial role in the rapid infiltration of rainfall to deep soil. Meanwhile, it also

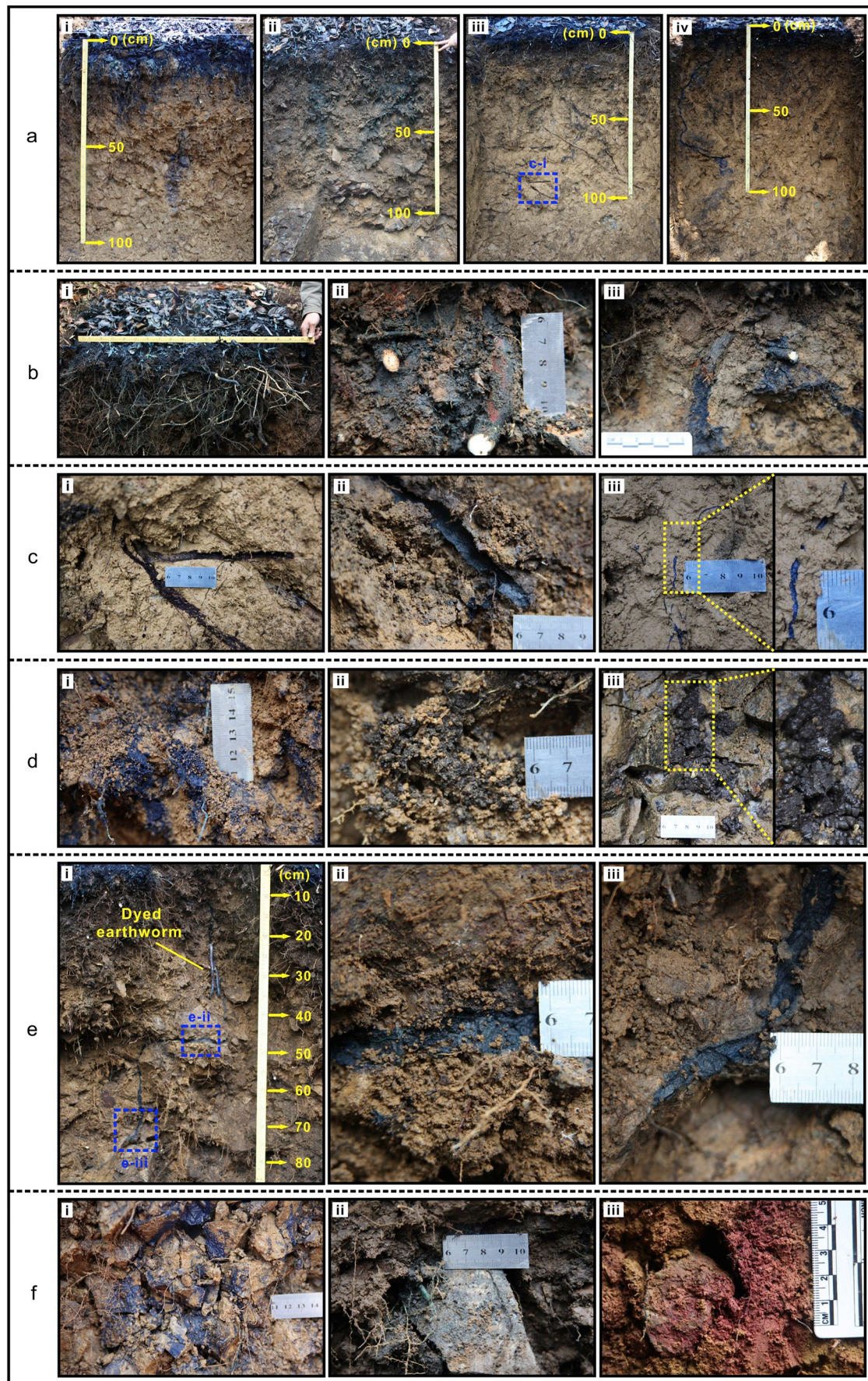


Fig. 13. Results of the dye tracer infiltration experiments. (a) is the typical dyed profiles, the ruler lengths are 1 m; (b)–(f) are the different types of macropores and some critical details.

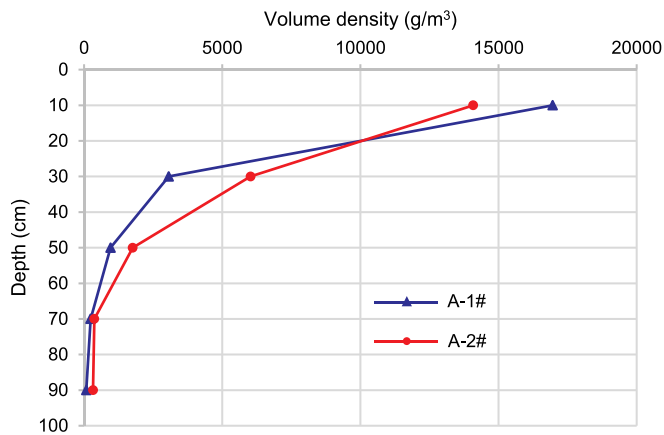


Fig. 14. Root volume densities at A-1# and A-1# sites plotted against depth (cm).

implies that water flows in them have changed into pipe flows, whose velocity is much faster than that of the flow in the porous matrix.

5.2.2.3. Inter-aggregate pores. Usually, fine earth contains abundant aggregates, and the pores between aggregates are defined as inter-aggregate macropores. Their dimensions are much larger than those of the pores between single particles, which largely improves soil permeability. Hence, when aggregates are present in soils, it is not the size of soil particles but the soil structures controlled by aggregates that determine soil permeability. Organic Matter (OM), produced by vegetation metabolism and litter degradation, is the basis for the formation of aggregates and resultant inter-aggregate macropores. Fig. 13d shows the dyed aggregates on the excavated profiles. Most of them occur above 30 cm, where vegetation roots producing OM massively develop (Fig. 14, Fig. 13d-i and ii).

Generally, OM is distributed in the range of upper 40 cm in soil profiles (Moore et al., 1986). However, some of aggregates were observed at about 1 m depth on some profiles where the bedrock emerged (Fig. 13d-iii). These aggregates presenting in the bedrock cracks should have been moved here by infiltration water, and rotten root channels formed by deep-rooted tree species may have facilitated the downward migration of the OM-rich aggregates by providing mobilization paths. Likewise, Noguchi et al. (1999) once indicated that some root channels were directly connected with cracks in the bedrock and water was observed flowing from these cracks during excavation. This deeper type of soil OM not only provides cements for aggregate development, but also supplements food for earth animals expanding their extent of activities. Through the above process, macropores in the deep connect with those in the subsurface soil, generating a united macropore network to promote the rapid infiltration of rainfall.

5.2.2.4. Worm pores. Burrowing animals, such as insects and earthworms, can form abundant macropores in forest soils. To gain access to food (OM), they exert repeated influences on soil structures, which will form preferential paths for water infiltration (Uchida et al., 2001). Fig. 13e shows a dyed earthworm passage of ~ 8 cm in diameter on an experimental profile. It extended to the depth of 75 cm and connected with the underlying bedrock cracks (similar phenomena can be found in Noguchi et al., 1999). Secretions generated by earthworms during their back and forth activities adhere to the passage walls, improving the water stability of soil structure. Once worm pores are formed, they can last for at least several hundred years (Green and Askeew, 1965; Ehlers, 1975; Omoti and Wild, 1979). Water could thus, like going through the macropores with respect to vegetation roots, move downward to groundwater tables through worm pore, without wetting the surrounding soil mass (Aubertin, 1971).

5.2.2.5. Earth-gravel interstices. Gravels of 4.75–75 mm (ASTM, 2017) diffusely distribute in the soil mass attached to the slopes, and their quantity has a tendency to increase with depth (Fig. 13a). The content of gravels may control the soil physical property, such as the bulk density and porosity, and then affect the soil hydraulic conductivity. Interstices between gravels and surrounding earth, here named the earth-gravel interstices, may form because of the erosive action of subsurface flows when the forces applied to individual soil particle by the flows exceed the structural competence of soils (Beven and Germann, 1982). Fig. 13f shows the dyed earth-gravel interstices found on some excavated profiles. Their sizes, with the width of 5–10 mm, were generally larger than those of inter-aggregate pores.

Because of the existence of various types of macropores, a mantle with strong capacities of absorbing and infiltrating is formed on the slopes. In particular, some macropores have extended to depth below 1 m (Fig. 13a-iii and iv). Hence, the macropore system developed in the slope unsaturated zone offers preferential paths for the infiltration of rainwater and determines the high infiltration rate of the slope soils. The rainwater can be simultaneously absorbed and rapidly infiltrate downward without overland flow generation. Out of these, the profile structure of soil layer plus bedrock in the Touzhai slopes is common, with the upper regolith of approximately 1–2 m in thickness and the underlying low permeability bedrock, which allow the infiltrated preferential flow to immediately arrive to the soil-bedrock interface in a short time and to turn to lateral flow to recharge the stream.

6. Conclusions

In the attempt to develop a cost-effective method of investigating the dynamics of the groundwater in vegetated slopes, the response behavior of the stream flow to rainfall in the vegetated Touzhai valley (Southwestern China) is examined based on in-situ monitoring as a first step. Afterwards combining the results from Complete Water Quality Analysis (CWQA), leaching experiment, field double-ring infiltration experiment as well as dye tracer experiment, the source of the stream water and the mechanisms of its response to rainfall are analyzed. The findings are summarized as follows.

- (1) The monthly, weekly and partial daily scales of stream Flow Rate (FR) and rainfall data indicate that the stream FR fluctuates sensitively with rainfall with surprisingly short lag times whether it is in the high- or low-water period. Therefore, the stream FR has a high indication precision for rainfall events.
- (2) The relatively high Total Dissolved Solid (TDS) and the similar magnitudes of SiO_2 concentrations in the stream water and leachate suggest that the stream water is almost entirely recharged by the groundwater in the valley slopes. Thus, the rapid variation of stream FR with rainfall reflects the real-time fluctuation of groundwater table following rainfall. The dynamics of the groundwater in vegetated slopes can be qualitatively estimated through stream FR.
- (3) The mechanism of the rapid response of groundwater to rainfall is clarified by the field experiments of double-ring infiltrometer and dye tracer infiltration. The mean saturated hydraulic conductivity (K_s) at the soil depths of 0, 10 and 60 cm are at a high level, with the values of 1.94×10^{-2} , 7.54×10^{-3} and 1.13×10^{-2} cm/s, respectively. What renders this result is the macropore system (comprised of root-soil interstices, rotten root channels, inter-aggregate pores, worm pores and earth-gravel interstices) well developed in the unsaturated zone of the slopes, which offers preferential paths for the infiltration of rainwater.

Thus, estimating the dynamics of the groundwater in vegetated slopes for evaluating the occurrence of landslides can be performed through monitoring of stream flows. More importantly, monitoring sites of stream flows may be placed at inhabited valley bottoms even on fan areas to avoid the lack of electric power and communication.

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